Aim: Color Image Segmentation using Image Processing Techniques

Objective: To develop and implement an algorithm for **color image segmentation** to partition an image into meaningful regions based on color information.

Software Requirements

- MATLAB / Python
- Libraries:
  - Open CV
  - NumPy
  - Matplotlib

Theory:

IMAGE SEGMENTATION refers to the process of partitioning a digital image into multiple segments (sets of pixels, also known as superpixels). Segmentation subdivides an image into its constituent regions or objects. The goal of segmentation is to simply and/or change the representation of an image into something that is more meaningful and easier to analyze. Image segmentation is typically used to locate objects and boundaries (lines, curves, etc.) in an image. More precisely, image segmentation is the process of assigning a label to every pixel with the same label share certain visual characteristics.

Segmentation algorithm for monochrome images generally based on one of two basic properties of image intensity values; discontinuity and similarity. In first category, the approach is to partition an image based on abrupt change in intensity; such as edge in an image. The principal approach in the second category are based on partitioning an image on regions that are similar according to a set of predefined criteria; examples are region splitting & merging , region growing, region splitting and merging

Mathematically, segmentation partitions an image III into regions R1,R2,...,Rn , such that:

- $\bigcup R_i = I$ (covers entire image)
- $R_i \cap R_j = \emptyset$(no overlap)

Each region is homogeneous based on certain features (color, intensity, texture)

Color Spaces Used

1. RGB Color Space

- Represents images using Red, Green, Blue components
- Sensitive to lighting variations

2. HSV (Hue, Saturation, Value)

- Hue represents color type
- More suitable for segmentation under varying illumination

3. Lab Color Space

- Separates luminance (L) from color components (a, b)
- Perceptually uniform

Methods of Color Image Segmentation

1. Thresholding-Based Segmentation

- Simple method using intensity/color thresholds
- Works well when object and background are distinct

2. K-Means Clustering (Unsupervised Method)

- Groups pixels into kkk clusters based on color similarity

3. Region-Based Segmentation

- Groups neighbouring pixels with similar properties
- Example: Region Growing

4. Edge-Based Segmentation

- Detects boundaries using gradients
- Less effective for color-only segmentation

- **Algorithm for Color Segmentation**

1. Input color image.
2. Pre-processing (noise removal if required).
3. Convert image to appropriate color space.
4. Apply segmentation method:
    - Thresholding / K-means / Region-based
5. Generate segmented output.
6. Display and analyse results.

Experimental Procedure

1. Load the input color image.
2. Convert RGB image to HSV or Lab color space.
3. Select segmentation method (e.g., K-means).
4. Apply clustering/thresholding.
5. Label segmented regions.
6. Display segmented output.
7. Compare results with original image.

Applications

- Autonomous vehicle vision systems
- Medical image analysis
- Object tracking
- Industrial quality inspection

Conclusion

This experiment demonstrates how **color information** can be effectively used to segment images. Among different techniques, **K-means clustering** provides flexible and efficient segmentation for multi-region images.

Questions

1. What is image segmentation?
2. Difference between RGB and HSV color space
3. Why is HSV preferred over RGB?
4. Explain K-means clustering algorithm
5. What are the limitations of thresholding?
6. Applications of color segmentation